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B.Tech. Degree III Semester Examination November 2017

CS 15-1306 DATA AND COMPUTER COMMUNICATION (2015 Scheme)

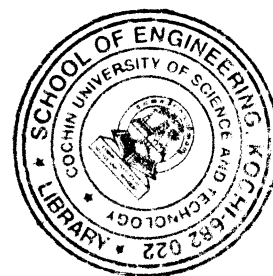
Time : 3 Hours

Maximum Marks : 60

PART A (Answer *ALL* questions)

(10 × 2 = 20)

- I. (a) Explain the key elements of a protocol.
- (b) What are the three criteria necessary for an effective and efficient network?
- (c) What are the three phases in a circuit switched network?
- (d) Briefly explain routing table in a datagram network.
- (e) Define piggybacking and its uses.
- (f) How does a single bit error differ from a burst error?
- (g) Explain different types of frequency based coding.
- (h) Explain the common fast Ethernet implementations.
- (i) Explain the function of repeater, hubs and routers.
- (j) Briefly explain the architecture of IEEE 802.11.



PART B

(4 × 10 = 40)

- II. (a) Explain the layers in the OSI model. (8)
- (b) The period of a signal is 100 ms. What is its frequency in Kilohertz? (2)
- OR**
- III. (a) Explain data rate limits. (8)
- (b) A periodic signal has a bandwidth of 20 Hz. The highest frequency is 60 Hz. What is the lowest frequency? Draw the spectrum if the signal contains all frequencies of the same amplitude. (2)
- IV. Explain different techniques used for multiplexing. (10)
- OR**
- V. (a) Explain the various modem standards. (5)
- (b) Describe the SS7 service and its relation to the telephone network. (5)
- VI. Define flow control. Explain the various ARQ methods. (10)
- OR**
- VII. (a) Explain CRC with an example. (8)
- (b) What is hamming distance? What is the minimum hamming distance? (2)
- VIII. Define network topology. Explain the different types of network topologies by mentioning its pros and cons. (10)
- OR**
- IX. Explain the different kinds of connecting devices used in the internet model. (10)